Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the OptiCrop system can be expanded to support more users, larger datasets, and additional agricultural features in future versions. The current implementation demonstrates a functional Flask-based crop recommendation system using machine learning, while future enhancements focus on improving scalability, automation, and intelligent decision support.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application runs primarily in a local environment.	Limited accessibility for remote users.	High
2	Weather data must be entered manually.	Reduces automation and user convenience	High
3	No fertilizer recommendation or disease detection features.	Limited agricultural decision support.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users.	Deploy on cloud platforms such as Render or PythonAnywhere.
2	Data Storage	Dataset stored as local CSV files.	Use a cloud database for larger agricultural datasets.
3	Performance	Single Flask server instance.	Use Gunicorn and load balancing for better scalability.
4	Security	Basic input validation implemented.	Add HTTPS, secure deployment, and enhanced validation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Weather API Integration	3–6 Months	Automatically fetch real-time weather data.
Phase 3	Fertilizer Recommendation System	6–9 Months	Suggest suitable fertilizers based on soil nutrients.
Phase 4	Crop Disease Detection and Yield Prediction	9–12 Months	Provide advanced agricultural decision support.